

Problem Statement 2 - Blue Coastal Action

Current State:

Nova Scotia has high and rapid tides, deep ports, high turbidity in our waters, and we are currently seeing an increase in tropical, invasive and unexpected species of fish that can be difficult to identify. Fish habitat monitoring can be done by capturing and handling fish, or passively with video surveys to avoid disturbing and stressing the animals. In most cases, fish handling is either prohibited or discouraged.

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Current issues around fish monitoring with video surveys include:

- Outdated species list, especially with climate change drivers.
- Low visibility and high interference from natural materials in the water (algae, sediment, grass, etc.)
- Reliance on human detection and associated errors and fatigue.
- Staff cost of analyzing video data.

We need a resource that can assist with fish detection, identification, and numeration within the conditions of Nova Scotia's waters.

Key Clients/ Audience:

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Limitations:

Due to environmental concerns, our use of AI is heavily restricted. At this time, we would not be able to support a publicly hosted AI model. However, we may be able to support an open-source model hosted locally, once we have the hardware to implement the system and more developed guidelines around the use of AI at Coastal Action.